Foundations of Artificial Intelligence - 2026.1

Sequential Decisions under Uncertainty - Competitive Environments

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Decision Making Situations

Decision Situation Matrix

Action Effects

Horizon		**Certainty**	**Risk**	**Uncertainty**
	Atomic	✓	✓	✓
	Sequential	✓	✓	NOW (33%)

Cooperative vs Competitive Situations

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Hence, we are more interested in a competitive scenario, where agents do not coordinate their actions.

Class Overview

1. A Ceteris Paribus Approach to Competitive Multi-Agent Situations
2. Alternating Markov Games
3. Exploration Based On Winning Probabilities

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But then what happens if, in the same game, we play against other policies $\pi'_2, .., \pi'_k$?

More reasonable task: Learn a policy π_1^* that is optimal in *all* possible MDPs.

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We will here assume that

- player 1 starts in a random initial state s_0
- rewards for player 1 only at the end, typically in $\{-1, 0, 1\}$
- $\gamma = 1$
- there is a known upper bound T on the maximum number of any trial

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Usually we do not have such strong opponent policies available to generate trials.

Reinforcement Learning for Competitive Multi-Agent Situations

If the assumption holds for all players, we can use a cross-MDP policy iteration idea:

Initialize $\pi_1, .., \pi_k$ randomly. Then, in an endless loop run

- Fix current player $i \leftarrow t \ \% \ k$
- Find π_i^* for the current MDP induced by $\pi_{\neq i}$
- Update $\pi_i \leftarrow \pi_i^*$
- If for k turns no policy has changed, stop

A fix point here would imply that we have found stable policies $\pi_1^*, .., \pi_k^*$.

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This approach would be typically *very* slow and learn a lot of irrelevant stuff ...

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Alternating Markov Games

An Alternating Markov Game is described as

$$\mathcal{G} = \langle S, \{A_i\}_{i=1}^k, \mathbb{P}, \{r_i\}_{i=1}^k, \gamma, \eta \rangle$$

where everything is as in a standard MDP except that

- A_i is the action space of player i,
- $r_i : S \to \mathbb{R}$ is the reward function of player i,
- $\eta : S \to \{1, \ldots, k\}$ determines the *active player* in each state.

At each step t:

1. The active player $i = \eta(s_t)$ selects an action $a_t \in A_i$ according to $\pi_i(a_t \mid s_t)$.
2. The environment transitions to $s_{t+1} \sim P(\cdot \mid s_t, a_t)$.
3. All players receive rewards $r_j(s_t)$.

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The trial for player i should

- only contain states $s_1^i, s_2^i, ..$ with $\eta(s_t^i) = i$, and
- as reward $r(s_t^i) := \sum_{s \in \tau[s_{t-1}^i : s_t^i]} r_i(s)$, thanks to $\gamma = 1$; rewards are now stochastic

Alternating Markov Games

Since we want to share knowledge across k GPI processes that would usually be generated in k independent PE steps, it is good to think of an alternative architecture.

One little invasive approach:

Make the PE routine a function of a trial τ (instead of conducting trials itself).
The PImp routine can stay the same.

Reinforcement Learning for 2-Player Games

Alternating Markov Games

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This has at least two effects

- Better usage of compute resources invested in trials:
  - Before: For 1 learning step of k agents we needed k trials.
  - Now: For 1 learning step of k agents we need 1 (shared) trial.
- Trials may implement updates in the opponents immediately:
  - Before: some number of trials for same opponent
  - Now: each trial can be based on updated opponents

Reinforcement Learning for 2-Player Games

Alternating Markov Games

There are special cases in which we can proceed slightly more efficient.

Consider a zero-sum game in which the only reward is in the last action.

In this case, we would learn a single q-function for both players simultaneously.

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In this case, we would learn a single q-function for both players simultaneously.

One can think of this literally as having a single agent that simultaneously plays both players.

Then we are (almost) back in the standard MDP case (but now with a state for each player).

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The only technical difficulty is that the reward at the end is only for one of the two players:

If it is 1 for player 1, the reward is -1 from the perspective of player 2.

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We somehow must take this into account when updating q-values.

The simplest way to do this is to multiply the previous iteration's utility with $-\gamma$ instead of γ, typically with -1.

Thereby, the bipolar aspect of the game is reflected in the q-updates.

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There is no point in learning *all* $q(s, a)$ values if most regions of the MDP are bad.

Then we must use our own policy for exploration.

Exploration Based On Winning Probabilities

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A state is relevant if it significantly influences $v(s_0)$ of an optimal policy.

So we might want to ignore states for which it is almost sure that they will not be visited by an optimal policy.

Unfortunately, we do not know this before.

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An optimal policy will seek to avoid states with low $v(s)$.

This gives a natural motivation to explore in such a way: Sample actions with a probability that is proportional to the current belief to win the game with that action.

$$\boxed{\pi(a|s) = \frac{\hat{q}(s, a)}{\sum_{a'} \hat{q}(s, a')}}$$

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2. it might not converge to a policy with (almost) probability 1 for a specific action.

Both issues can be solved by picking actions that maximize the UCB score instead:

$$\boxed{u(s,a) = \hat{q}(s,a) + \sqrt{\frac{\log \sum_{a'} N[s,a']}{N[s,a]}}}$$

This is not randomly picking yet exploring forever while converging.

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The problem is now that PE does not evaluate π anymore but π'.

This can break the assumption that PE makes $\hat{q}^{\pi}$ converge to q^{π} (because we get $\hat{q}^{\pi'}$).

However, if π greedily follows $\arg\max_a \hat{q}(s, a)$ and π' follows $\arg\max_a u(s, a)$, then these policies are almost always identical for sufficiently large sample sizes.

Summary

Main takeaways:

- Competitive Multi-Agent Environments require a special treatment
- A rather efficient approach is to generate shared trials and synchronously update policies across players
- In this case, we cannot use Exploring Starts anymore but can control exploration via the policy.
- Even though we now have an exploring policy, $\hat{q}$ converges to the q in the maximally competitive scenario.